

8(a)	<i>max 4 of:</i> 1 different <u>habitats</u> ; 2 different <u>niches</u> ; 3 many (different) <u>species</u> / large variety of <u>species</u> ; 4 <i>ref. to</i> (much) genetic diversity within a species ; 5 different <u>selection</u> pressures ; 6 <i>ref. to</i> <u>adaptation</u> ; 7 different, climate / rainfall / temperature / soil / topography / conditions ;	4
8(b)(i)	both sites are the same / no (significant) difference between two sites ;	1
8(b)(ii)	<i>genera 2 and species 4 ;</i>	1

Question	Answer	Marks																								
8(b)(iii)	<i>all figures to 3 d.p. to score but only penalise extra d.p. or rounding error associated with extra d.p. once</i> <table><tr><th>species</th><th>number on grassland not grazed</th><th>n / N</th><th>$(n / N)^2$</th></tr><tr><td><i>Onthophagus pennsylvanicus</i></td><td>6641</td><td><u>0.873</u></td><td>0.762</td></tr><tr><td><i>Canthon ebenus</i></td><td>774</td><td><u>0.102</u></td><td>0.010</td></tr><tr><td><i>Canthon pilularius</i></td><td>108</td><td><u>0.014</u></td><td>0.000</td></tr><tr><td><i>Onthophagus hecate</i></td><td>85</td><td><u>0.011</u></td><td>0.000</td></tr><tr><td>total</td><td>7608</td><td></td><td>0.772</td></tr></table> n / N figures correct / numbers of each species divided by total ; $(n / N)^2$ calculated and added up ; ecf from incorrect column 1 including figures with fewer / more than 3 d.p. 0.228 ; ecf total figure subtracted from 1	species	number on grassland not grazed	n / N	$(n / N)^2$	<i>Onthophagus pennsylvanicus</i>	6641	<u>0.873</u>	0.762	<i>Canthon ebenus</i>	774	<u>0.102</u>	0.010	<i>Canthon pilularius</i>	108	<u>0.014</u>	0.000	<i>Onthophagus hecate</i>	85	<u>0.011</u>	0.000	total	7608		0.772	3
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8(b)(iv)	greater species <u>evenness</u> on grazed grassland ; ora A mostly, one species / <i>O. pennsylvanicus</i>, on not grazed grazing increases (dung beetle species) (bio)<u>diversity</u> ; ora <i>if opposite conclusion reached check answer for (iii) and apply ecf for mp2 if $D > 0.521$</i>	2																								